

# The Two Assignments Most of You Missed

Re-issued 6 October 2026 · the Library task from 26 September and the Missing Fest Trophy from 30 September, written out in full

A lot of you told me you did not know what these two were. That is fair — both were given verbally in class and never written down. That was my mistake, not yours. Here they are properly. Nothing in either one goes beyond Session 6, 7 and 8.

## 1. The Library task

26 SEPTEMBER · GIVEN IN CLASS AS HOMEWORK · 3 OF 26 SUBMITTED IT

This was the **second** task of that session. The first one, in class, was the `Bank` class with `deposit`, `withdraw` and `get_balance`. Most of you did that one and stopped there — which is why a lot of you believe you submitted this. You submitted the Bank. The Library was the homework.

### What was asked

Two classes.

**Book** — holds one title.

- A constructor taking the title, the author and the number of copies.
- The copies stored as a **private** attribute, with two underscores.
- A `__str__` so that printing a book shows something readable instead of `<__main__.Book object at 0x...>`.

**Library** — holds many books. Three things it must do:

1. **Add a book.** If the title is already in the library, raise its copy count instead of creating a second entry for the same book.
2. **Show all books.** Print every title with how many copies are available.
3. **Borrow a book.** Reduce the available count by one. If there are no copies left, refuse it and say so. The count must never go below zero.

If you finished those three, the extra one was **return a book** — add a copy back, and refuse the return if nothing was out on loan in the first place.

### The three traps

| TRAP                                  | WHAT GOES WRONG                                                                                                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counting copies as one number         | You need two ideas: how many the library owns, and how many are on the shelf right now. One number cannot tell you both.                                                           |
| Duplicating a title                   | Adding "Ikigai" twice should give you one book with 2 copies, not two books with 1 copy each. Search before you append.                                                            |
| Using Book objects as dictionary keys | Two Book objects with the same title are not equal as far as Python is concerned, so <code>if book in self.books</code> will always be False. Compare the titles, not the objects. |

**You do not need to do this one separately any more.** Assignment 6 — the GitHub one issued today — is this task, written out task by task with a marker attached. Do Assignment 6 and this is covered.

## 2. The Missing Fest Trophy

30 SEPTEMBER · SET PERSONALLY · 1 OF 26 SUBMITTED IT

This one is a small text adventure. The point of it is not the story — it is that you cannot write it

without objects. A player, a place and a chest each have their own state, and the whole thing only works when they talk to each other.

## The story

The trophy for the college fest has gone missing from the main stage. You have to find it. There are three places you can go.

- **The canteen** — there is a key hidden under one of the tables.
- **The computer lab** — it is locked. You need the key to get in. Inside, a secret code is written on one of the computers.
- **The main stage** — the trophy chest is here. It only opens with the code.

## What was asked

At least three classes.

- **Player** — has a name and a bag. `collect(item)` puts something in the bag, and refuses if it is already there. `show_bag()` prints what you are carrying, or says the bag is empty.
- **Place** — has a name and a description. `visit()` prints them. A locked place also knows which item is needed to open it, and refuses to let you in until you have it.
- **Chest** — knows the code it wants. `open(player)` checks whether the player is carrying that code. If yes, the trophy is found. If no, it stays shut.

## The part that carried the marks

At the bottom of your file you had to run the story **twice** and print both:

```
===== FAILED ROUTE =====
Go straight to the stage. Try the chest. It stays locked.
Try the lab. It stays locked. Your bag is empty.

===== WINNING ROUTE =====
Canteen -> take the key -> lab -> unlock it -> take the code
-> stage -> open the chest -> trophy found.
```

Showing the failed route first is the whole exercise. It proves your objects actually hold state — that the lab is genuinely locked until the key is in the bag, rather than you just printing the happy ending in order.

**This one stays optional.** It is the most enjoyable thing in the module and it is the best preparation for the agent work in Module 3, where an agent carries state and decides what it can and cannot do next. If you have time after Assignment 6, do it. Upload it into your Drive folder as before.